

## JAVA ARRAY & STRING PRACTICE SHEET (CONSTRAINED)

Rules:

1. No StringBuilder or StringBuffer.
2. Allowed String methods: charAt(), split(), length(), and toCharArray().
3. Language: Java

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--- BEGINNER LEVEL ---

### 1. Sum of Elements

Calculate the total sum of an integer array.

- Signature: public int getSum(int[] nums)
- Input: [1, 2, 3, 4] | Output: 10

### 2. Find Maximum

Find the largest integer in an unsorted array.

- Signature: public int findMax(int[] nums)
- Input: [5, 2, 9, 1, 6] | Output: 9

### 3. Count Occurrences

Count how many times a target integer 'x' appears in the array.

- Signature: public int countX(int[] nums, int x)
- Input: nums = [1, 2, 1, 3, 1], x = 1 | Output: 3

### 4. Reverse an Array

Reverse the elements of an array in-place.

- Signature: public void reverse(int[] nums)
- Input: [10, 20, 30] | Output: [30, 20, 10]

### 5. Linear Search

Return the index of the first occurrence of 'target'. Return -1 if not found.

- Signature: public int search(int[] nums, int target)
- Input: nums = [4, 8, 2], target = 8 | Output: 1

### 6. Reverse a String (Manual)

Reverse a string by creating a char[] and swapping elements manually.

- Signature: `public String reverseString(String s)`
- Input: "java" | Output: "avaj"

#### 7. Vowel Counter

Count 'a', 'e', 'i', 'o', 'u' using `charAt()`.

- Signature: `public int countVowels(String s)`
- Input: "Hello World" | Output: 3

#### 8. Palindrome Check

Compare characters from start and end using `charAt()`.

- Signature: `public boolean isPalindrome(String s)`
- Input: "racecar" | Output: true

#### 9. Remove Whitespace

Use `split(" ")` and concatenate the resulting parts back together.

- Signature: `public String removeSpaces(String s)`
- Input: " a b c " | Output: "abc"

#### 10. Toggle Case

Convert lowercase to uppercase (ASCII math: `'a' - 32 = 'A'`) and vice versa.

- Signature: `public String toggleCase(String s)`
- Input: "aBc" | Output: "AbC"

### --- INTERMEDIATE LEVEL ---

#### 11. Two Sum

Find the indices of two numbers that add up to a target.

- Signature: `public int[] twoSum(int[] nums, int target)`
- Input: `nums = [2, 7, 11]`, `target = 9` | Output: `[0, 1]`

#### 12. Remove Duplicates (Sorted Array)

Remove duplicates in-place (don't worry about array size, just move values).

- Signature: `public int removeDuplicates(int[] nums)`
- Input: `[1, 1, 2]` | Output: 2 (Array starts with `[1, 2]`)

#### 13. Move Zeroes

Move all 0's to the end without changing the order of non-zero elements.

- Signature: `public void moveZeroes(int[] nums)`
- Input: `[0, 1, 0, 3, 12]` | Output: `[1, 3, 12, 0, 0]`

#### 14. Rotate Array

Rotate the array to the right by `k` steps.

- Signature: `public void rotate(int[] nums, int k)`
- Input: `[1, 2, 3, 4, 5]`, `k = 2` | Output: `[4, 5, 1, 2, 3]`

#### 15. Valid Anagram

Check if two strings are anagrams using a frequency array of size 26.

- Signature: `public boolean isAnagram(String s, String t)`
- Input: `s = "anagram"`, `t = "nagaram"` | Output: `true`

#### 16. First Unique Character

Find the index of the first non-repeating character using `charAt()`.

- Signature: `public int firstUniqChar(String s)`
- Input: `"leetcode"` | Output: `0`

#### 17. String Compression (Logic-heavy)

Compress `"aaabb"` to `"a3b2"`. Since you can't use `StringBuilder`, use a `char[]`.

- Signature: `public String compress(String s)`
- Input: `"aabbcc"` | Output: `"a2b2c2"`

#### 18. Longest Common Prefix

Find the longest common prefix among an array of strings.

- Signature: `public String longestCommonPrefix(String[] strs)`
- Input: `["flower", "flow", "flight"]` | Output: `"fl"`

--- ADVANCED LEVEL ---

#### 19. Sliding Window Maximum

Find the max element in every sliding window of size `k`.

- Signature: `public int[] maxSlidingWindow(int[] nums, int k)`
- Input: `nums = [1, 3, -1, -3, 5]`, `k = 3` | Output: `[3, 3, 5]`

#### 20. Longest Substring Without Repeating Characters

Find the length of the longest substring without duplicates using a sliding window.

- Signature: `public int lengthOfLongestSubstring(String s)`
- Input: "abcabcbb" | Output: 3

#### 21. Trapping Rain Water

Calculate trapped water. Hint: Use two pointers or prefix/suffix arrays.

- Signature: `public int trap(int[] height)`
- Input: [0, 1, 0, 2, 1, 0, 1, 3] | Output: 6

#### 22. Minimum Window Substring

Smallest substring of S containing all characters of T. Use `charAt()` extensively.

- Signature: `public String minWindow(String s, String t)`
- Input: s = "ADOBECODEBANC", t = "ABC" | Output: "BANC"